Problem

Title

Armstrong Numbers in a Given Range

Topics

Number Mathematics, Digit Extraction, Power Calculation, Loops, Nested Loops, Conditional Statements

Difficulty Level

Medium

Problem Detailed Statement (Scenario Based)

In a number theory workshop, students are introduced to a special category of numbers known as **Armstrong Numbers** (also called Narcissistic Numbers).

An Armstrong number is defined as a number that is equal to the sum of its digits each raised to the power of the total number of digits.

For example:

- 153 → It has 3 digits
  ($1^3 + 5^3 + 3^3 = 1 + 125 + 27 = 153$) ✓
- 9474 → It has 4 digits
  ($9^4 + 4^4 + 7^4 + 4^4 = 9474$) ✓

Now, the trainer provides two integers **L** and **R**, defining a range. Students must:

1. Iterate through all numbers from L to R.
2. For each number:
   - Count its digits.
   - Extract each digit using loop logic.
   - Raise each digit to the power of total digits.
   - Compute the sum.
3. Check whether the computed sum equals the original number.
4. Print all Armstrong numbers within the range.

Students are not allowed to use:

- Built-in power functions directly (optional restriction for deeper practice).
- String conversion methods for digit extraction.

They must rely on:

- Mathematical operations (% and /)
- Nested loops
- Proper conditional checking

This problem enhances:

- Digit manipulation skills
- Mathematical reasoning
- Nested loop implementation
- Edge case handling

If no Armstrong number exists in the range, print `-1`.

Input Format

Two space-separated integers:

```
L R
```

Constraints:

```
0 ≤ L ≤ R ≤ 10^6
```

Output Format

- Print Armstrong numbers in ascending order separated by space.
- If none exist, print:

```
-1
```

Examples

Example 1

Input:

```
1 200
```

Output:

```
1 153
```

Example 2

Input:

```
150 160
```

Output:

```
153
```

Example 3

Input:

```
10 99
```

Output:

```
-1
```

Example 4

Input:

```
9000 9500
```

Output:

```
9474
```